Survey overview

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Summary

We surveyed 80 CFs across eight provinces and conducted 549 interviews in total. The sample included CFs from ~20 ha to >80,000 ha and that varied in years since establishment from one to 7 yrs. In each community forest, we conducted four FGDs with (i) CFMG executive, (ii) community men, (iii) community women, and (iv) community youths. This enables us to examine differences in perspectives towards CFM across these stakeholder groups. In addition, we surveyed individual key stakeholders, including DFOs, CFMG executive members (usually the Secretary), NGO or private sector partners, and traditional leaders. Overall, the sample was male biased, reflecting gender biases in roles at the community level, but our information is balanced through specifically including a female FGD in each community. The sample included a wide range of ages from ~20 to >80 years. Although most interviewees had lived all their lives in the community, the sample include some who had moved into the community recently. Our sampling approach ensured an unbiased selection and these data demonstrate that we covered a wide range of CFs, in terms of geography, size and age, and interviewees, in terms of gender, age and roles. Hence, our results may be considered generalisable at the national level.

Community Forests surveyed

The sampling protocol targeted 80 CFs across eight provinces. Initially we attempted to sample ten CFs per province. However, as some provinces had very few CFs, this was not possible. Hence, to make up the numbers, we selected up to 14 CFs from provinces with more.

We were unable to collect data on five of the 80 CFs, because the Community Forest Management Group (CFMG) was defunct. The final sample for analysis, therefore, included 75 CFs. The list of surveyed CFs is given in Appendix 2.

Number of Community Forests surveyed

The numbers per province are given in Figure 1. As can be seen, Southern province have the fewest, where we were only able to survey five CFs, and Muchinga had the most with 13 surveyed CFs.

```r
cfmg |> filter(!is.na(Name)) |>
        group_by(province,Name) |>
        summarise(
                Obs = sign(n()),
                .groups = "drop"
        ) |>
        group_by(province) |>
        summarise(
                Obs = sum(Obs)
        ) |>
        ggplot(aes(x = province, y = Obs)) +
        geom_col(position = position_dodge(width = 0.9), color = "black") +
        theme_bw(base_size = 14) +
        theme(legend.position = "none",
              axis.text.x = element_text(size = rel(1.14), angle = 45,
                                          hjust = 1, vjust = 1),
              axis.text.y = element_text(size = rel(1.14)),
              axis.title = element_text(size = rel(1.29)),
              ) +
        ylim(0,15) +
        xlab("") +
        ylab("Number of Community Forests")
```

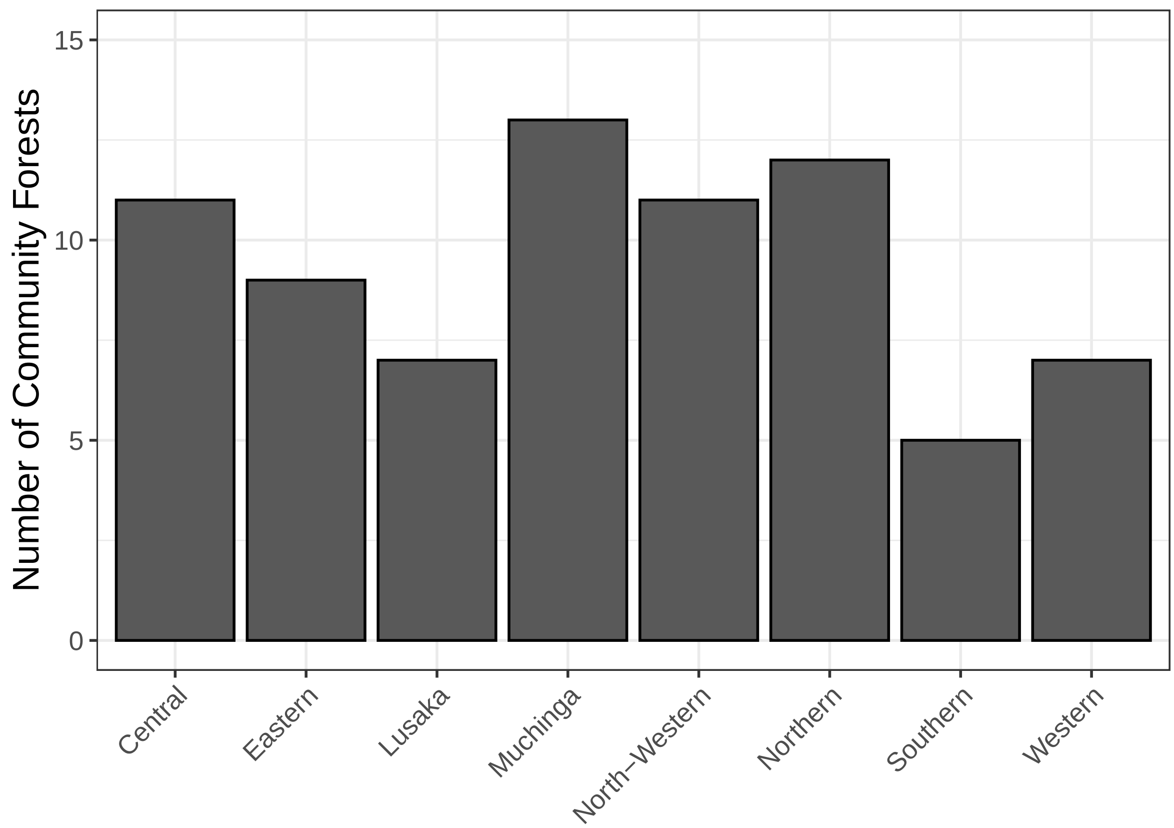

Figure 1: Number of community forests surveyed by province.

Size of community forests surveyed

The CFs surveyed range widely in size from less than 100 ha to over 100,000 ha (Figure 2). Within provinces, we surveyed a wide range of CF sizes, but the median sizes varied from around 900 ha in Muchinga Province to approximately 80,000 ha in Lusaka Province.

```
cfmg |> group_by(province,Name) |>
        summarise(
                Size = mean(`Size (Hectares)`, na.rm = TRUE),
                .groups = "drop"
        ) |>
        ggplot(aes(x = province, y = Size)) +
        geom_boxplot() +
        geom_jitter()+
```

```
    scale_y_log10() +
    stat_summary(fun=mean, geom="point", size=3, col = 'red') +
    theme_bw(base_size = 14) +
    theme(legend.position = "none",
          axis.text.x = element_text(size = rel(1.14), angle = 45,
                                     hjust = 1, vjust = 1),
          axis.text.y = element_text(size = rel(1.14)),
          axis.title = element_text(size = rel(1.29)),
          ) +
    #ylim(0,15) +
    xlab("") +
    ylab("Size of Community Forests")
```

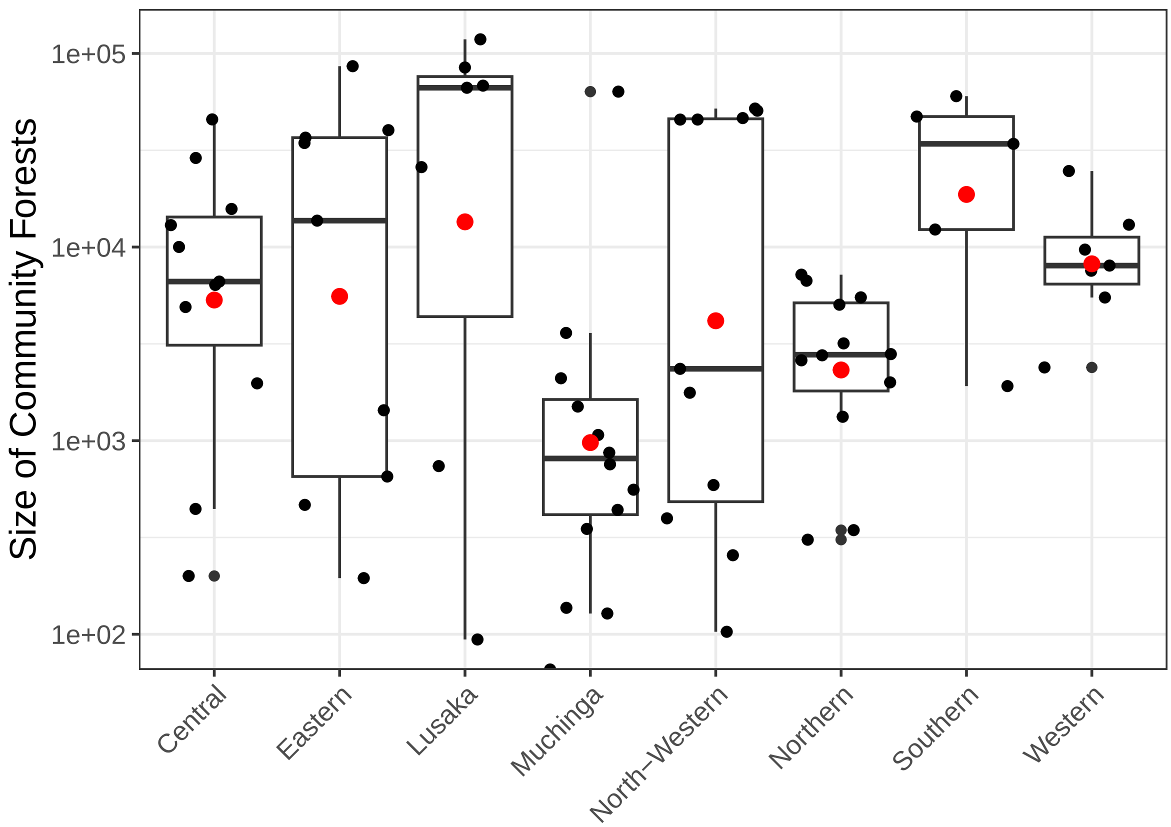

Figure 2: Size of community forests surveyed by province. The black dots represent the individual CFs. The boxes represent the 25th to 75th percentiles, with the black bar denoting the 50th percentile. The red dot indicates the mean size.

Through the survey tool, we asked respondents the size of their community forest (CF). Using only responses from the CFMG executive groups, their answers are plotted against the size given in the FD database (Figure 3). We restricted answers to the CFMG executive group, as we expected them to be the best informed as the the official size of their CF.

```r
cfmg |> filter(interview_type == "cfmg_executive") |>

        ggplot(aes(x = cfmg_governance_size_cfma,
                   y = `Size (Hectares)`)) +
        geom_abline(intercept = log(1), slope = 1,color = "lightblue",
              linewidth = 1.2) +
        geom_point(alpha = 0.6) +
        theme_bw(base_size = 14) +
        theme(legend.position = "none",
              axis.text.x = element_text(
                      size = rel(1.14), angle = 0, hjust=0.5),
              axis.text.y = element_text(size = rel(1.14)),
              axis.title.x = element_text(size = rel(1.29)),
              axis.title.y = element_text(size = rel(1.29))) +
        scale_y_log10(
                limits = c(10,100000)) +
        scale_x_log10(
                limits = c(10,100000)
        ) +
        xlab("Reported size (ha)") +
        ylab("Size (ha) in dataset")
```

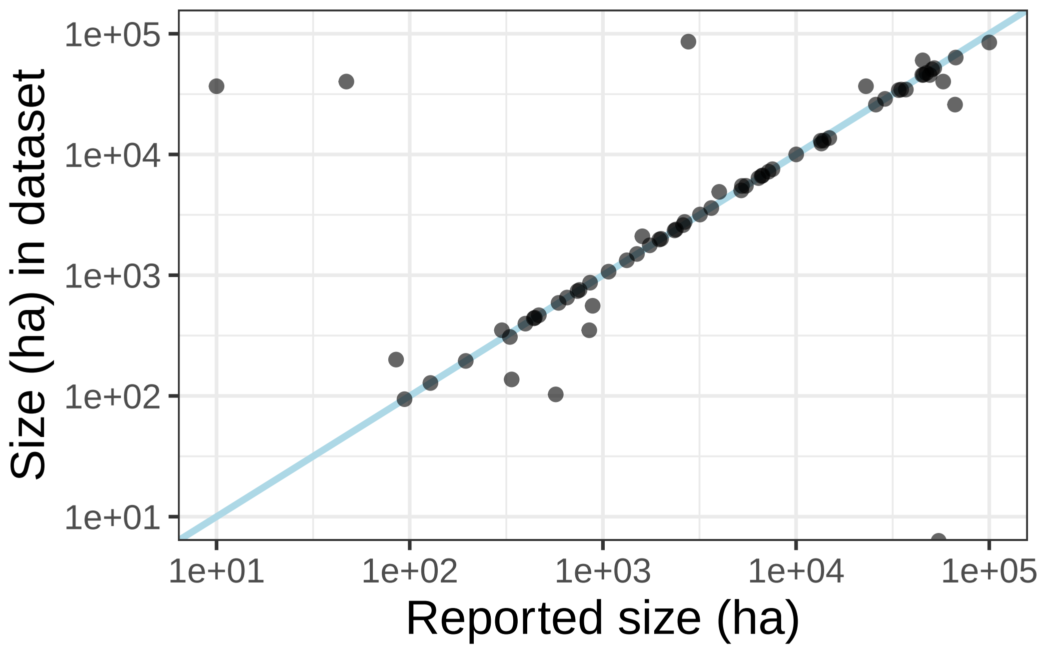

Figure 3: Size of CF as reported by the CFMG executive against the size recorded in the Forest Department database

While most answers lie along the line, indicating perfect agreement between the FD database and the responses, several are off - some by very large amounts (note that the axes are log-scale). In some cases, these differences may reflect errors in the FD database. However, the other possibility is that the communities are misinformed as to the size of their CF, which in turn suggests Free, Prior and Informed Consent (FPIC) may not have been properly conducted.

Age of community forests

The regulations enabling the formation of CFs were approved in 2018. Hence, the oldest CFs are approximately 8 yrs. However, most are less than 3 yrs old. We randomly selected CFs from among those that were at least 1 yr old in Jan 2024 (Figure 4). However, the data on Establishment Date in the FD database has some errors, so some CFs appear younger.

```
cfmg |> group_by(province,Name) |>
        summarise(
                Mean_date = mean(`Date Established`),
                .groups = "drop"
        ) |>
```

```
mutate(
        Age = interval(Mean_date,ymd("2025-01-01")) / dyears(1),
) |>
ggplot(aes(x = province, y = Age)) +
geom_boxplot() +
geom_jitter()+
scale_y_continuous(limits = c(0,8)) +
stat_summary(fun=mean, geom="point", size=3, col = 'red') +
theme_bw(base_size = 14) +
theme(legend.position = "none",
      axis.text.x = element_text(size = rel(1.14), angle = 45,
                                 hjust = 1, vjust = 1),
      axis.text.y = element_text(size = rel(1.14)),
      axis.title = element_text(size = rel(1.29)),
      ) +
xlab("") +
ylab("Age of Community Forests on 01/01/2025")
```

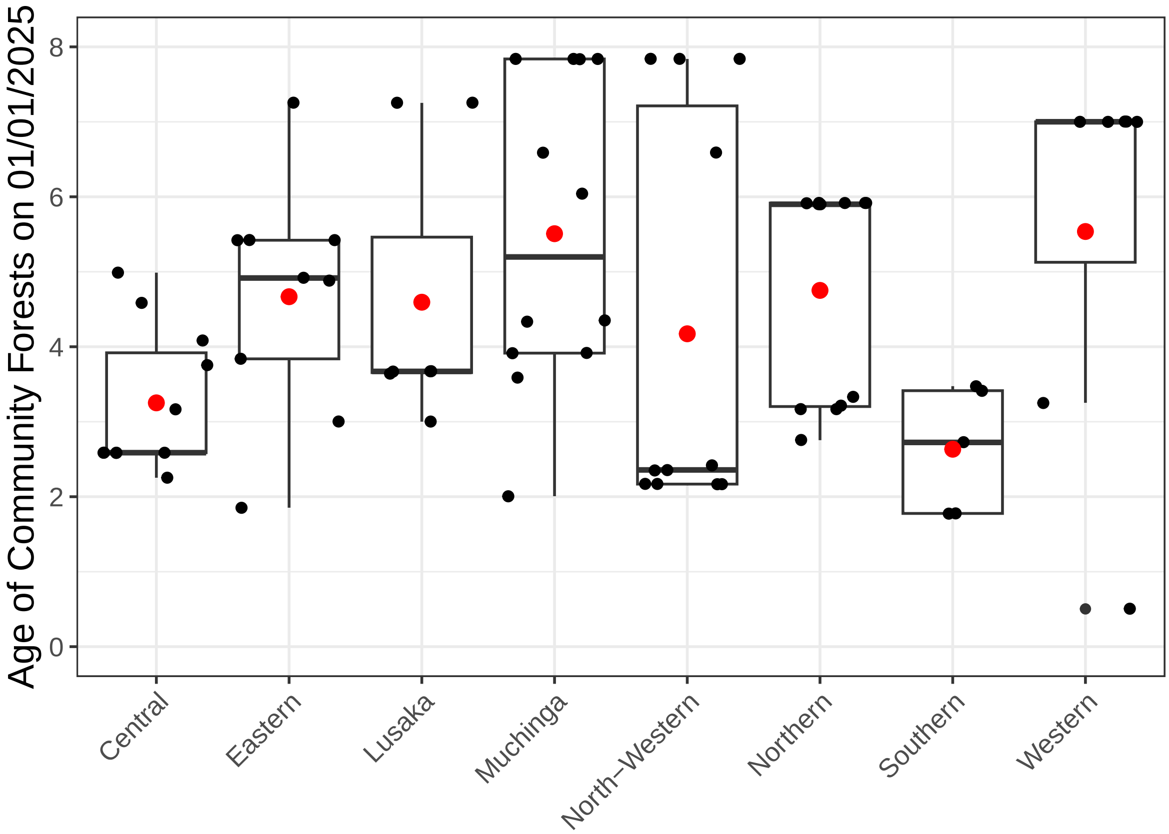

Figure 4: Age of community forests surveyed by province. The black dots indicate the individual CFs. The boxes represent the 25th and 75th percentile and the 50th percentile is indicated with the black bar. The red dot indicates the mean.

Interviews

Number of interviews

A total of 549 interviews across 75 Community Forest Management Groups (CFMGs) were surveyed across the eight provinces. The number of interviews per CFMG varied from five to 14 (Figure 5).

```
cfmg |> group_by(province,cfmg_name) |>
        summarise(
                Obs = n(),
                .groups = "drop"
```

```
) |>
group_by(province) |>
summarise(
        n_int = mean(Obs),
        sd_int = sd(Obs),
        .groups = "drop"
) |>
ggplot(aes(x = province, y = n_int)) +
geom_col(position = position_dodge(width = 0.9), color = "black") +
geom_errorbar(aes(ymin = n_int - sd_int, ymax = n_int + sd_int),
              position = position_dodge(width = 0.9),
              width    = 0.25,
              linewidth = 0.8
              ) +
theme_bw(base_size = 14) +
theme(legend.position = "none",
      axis.text.x = element_text(size = rel(1.14), angle = 45,
                                 hjust = 1, vjust = 1),
      axis.text.y = element_text(size = rel(1.14)),
      axis.title = element_text(size = rel(1.29)),
      ) +
ylim(0,15) +
xlab("") +
ylab("Number of interviews")
```

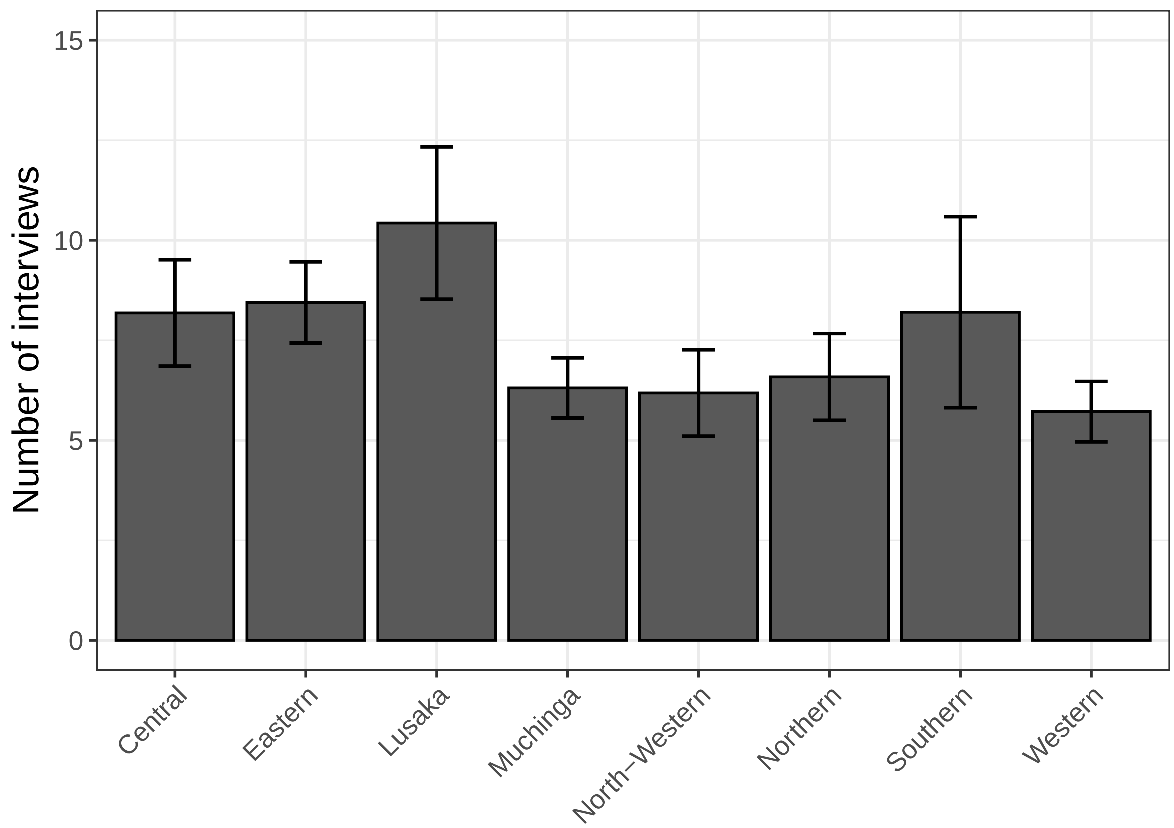

Figure 5: Number of interviews per CF. Mean (main bar) and standard deviation (whisker) of the number of interviews held per CFMG by province

In every CFMG, we aimed to hold four FGDs including, (i) CFMG executive, (ii) community members - male, (iii) community members - female, and (iv) community members - youth. Below (Table 1) are the total numbers of the different categories of FDG we were able to conduct. As can be seen, we were unable to arrange FGDs for community males and youths in some CFMGs. Meanwhile, we have an extra FGD for female community members, because in one CF two groups were surveyed.

```
cfmg |> filter(interview_type != "individual_interview") |>
        droplevels() |>
        group_by(interview_type) |>
        summarise(
                n = n()
        ) |>
        mutate(
```

```r
                interview_type = case_when(
                        interview_type == "cfmg_executive" ~  "CFMG Executive",
                        interview_type == "cfmg_female" ~  "Community - Female",
                        interview_type == "cfmg_male" ~  "Community - Male",
                        interview_type == "cfmg_youth" ~  "Community - Youth",
                        TRUE                              ~ interview_type
                )
        ) |>
        gt(rowname_col = "interview_type") |>
        tab_stubhead(label = "FDG type") |>
        tab_header(title = "FGD groups by type",
                   subtitle = "n = number of FGD groups of each type.") |>
        opt_align_table_header(align = "left") |>
        cols_align(align = "center", columns = everything()) |>
        tab_style(
          style     = cell_text(align = "left"),
          locations = cells_stub()
        ) |>
        tab_style(
          style     = cell_text(align = "center"),
          locations = cells_stubhead()
        ) |>
        sub_missing()
```

Table 1: Number of Focal Group Discussions by type

FGD groups by type

n = number of FGD groups of each type.

FDG type	n
CFMG Executive	75
Community - Female	76
Community - Male	72
Community - Youth	69

The number of individual interviews with key stakeholders also varied among the CFs. However, we attempted to hold interviews with a CFMG executive member (usually the secretary), the DFO and the community headmen or headwoman in every CF (Table 2).

```r
cfmg |> filter(!is.na(interviewee_role)) |>
        group_by(interviewee_role) |>
        summarise(
                n = n()
        ) |>
        mutate(
                interviewee_role = case_when(
                        interviewee_role == "cfmg_exec_member" ~  "CFMG Executive",
                        interviewee_role == "cfmg_member" ~  "Community member",
                        interviewee_role == "trad_leader" ~  "Traditional leader",
                        interviewee_role == "dfo" ~  "District Forest Officer",
                        interviewee_role == "NGO_rep" ~  "NGO representative",
                        TRUE                                  ~ interviewee_role)
                ) |>
        gt(rowname_col = "interviewee_role") |>
        tab_stubhead(label = "Interviewee Role") |>
        tab_header(title = "Individual interviews by interviewee role",
                   subtitle = "n = number of individual interviews by interviewee role.") |>
        opt_align_table_header(align = "left") |>
        cols_align(align = "center", columns = everything()) |>
        tab_style(
          style     = cell_text(align = "left"),
          locations = cells_stub()
        ) |>
        tab_style(
          style     = cell_text(align = "center"),
          locations = cells_stubhead()
        ) |>
        sub_missing()
```

Table 2: Number of Individual Interviews by interviewee role

Individual interviews by interviewee role
n = number of individual interviews by interviewee role.

Interviewee Role	n
Community member	31
CFMG Executive	63
Traditional leader	75
District Forest Officer	56
NGO representative	32

Gender breakdown of interviewees

Reflecting gender biases among the stakeholder groups, there was a much higher representation of men than of women among the key stakeholders interviewed (Table 3, Figure 6).

```
cfmg |> filter(interview_type != "individual_interview") |>
        group_by(interview_type) |>
        summarise(
                senior_male = mean(senior_male),
                senior_female = mean(senior_female),
                youth_male = mean(youth_male),
                youth_female = mean(youth_female)
        ) |>
        mutate(
                interview_type = case_when(
                        interview_type == "cfmg_executive" ~  "CFMG Executive",
                        interview_type == "cfmg_female" ~  "Community - Female",
                        interview_type == "cfmg_male" ~  "Community - Male",
                        interview_type == "cfmg_youth" ~  "Community - Youth",
                        TRUE                               ~ interview_type
                )
        ) |>
        gt(
                rowname_col = "interview_type"
        ) |>
        tab_stubhead(label = "FDG Type") |>
        cols_label(
                senior_male  = "Senior Male",
                senior_female = "Senior Female",
                youth_male   = "Youth Male",
                youth_female = "Youth Female"
        ) |>
        fmt_number(columns = c(senior_male, senior_female, youth_male, youth_female), decima
        tab_header(title = "Gender breakdown among FGD participants",
                   subtitle = "Values represent the mean number of participants of each gende
        opt_align_table_header(align = "left") |>
        cols_align(align = "center", columns = everything()) |>
        tab_style(
          style     = cell_text(align = "left"),
          locations = cells_stub()
        ) |>
        tab_style(
          style     = cell_text(align = "center"),
```

```
          locations = cells_stubhead()
        ) |>
        sub_missing()
```

Table 3: Gender and youth breakdown of FGDs. Mean number of participants in each category.

Gender breakdown among FGD participants

Values represent the mean number of participants of each gender/age group per FGD type

FDG Type	Senior Male	Senior Female	Youth Male	Youth Female
CFMG Executive	3.5	1.7	1.0	0.0
Community - Female	0.0	5.9	0.0	0.0
Community - Male	6.6	0.0	0.0	0.0
Community - Youth	0.0	0.0	4.5	3.4

```
cfmg |> filter(interview_type == "individual_interview") |>
        group_by(interviewee_role,demo_information_interviewee_gender) |>
        summarise(
                Obs = n()
        ) |>
        rename(
                Gender = demo_information_interviewee_gender
        ) |>
        mutate(
    Gender = tolower(Gender),
    Obs_plot = if_else(Gender == "female", -Obs, Obs),
    interviewee_role = case_when(
            interviewee_role == "trad_leader" ~ "Traditional leader",
            interviewee_role == "NGO_rep" ~ "NGO representative",
            interviewee_role == "dfo" ~ "District Forest Officer",
            interviewee_role == "cfmg_member" ~ "Community member",
            interviewee_role == "cfmg_exec_member" ~ "CFMG executive",
    )
  ) |>
        ggplot(aes(x = interviewee_role, y = Obs_plot, fill = Gender)) +
        geom_col(position = "identity") +
        coord_flip() +
        scale_y_continuous(
                breaks = seq(-20, 50, by = 10),
                labels = abs,
                limits = c(-20,50)
```

```
            ) +
      scale_fill_manual(
            values = c("female" = "#FFAB91", "male" = "#80CBC4"),
            labels = c("Female", "Male")
            ) +
      geom_hline(yintercept = 0, colour = "white", linewidth = 1.2) +
      geom_vline(xintercept = 0, color = "grey30", linewidth = 0.7) +

labs(
        x = NULL,
        y = NULL) +
theme_minimal(base_size = 14) +
      theme(
            legend.position = "bottom",
            legend.key.justification = "centre",

  # Remove all vertical grid lines except the zero line (already kept via geom_vline)
  panel.grid.major.x = element_blank(),
  panel.grid.minor.x = element_blank(),
  panel.grid.major.y = element_blank(),
  panel.grid.minor.y = element_blank(),

  axis.text.y = element_text(size = rel(1.14), vjust = 0.5),
  axis.text.x = element_text(size = rel(1.14)),
  plot.margin = margin(t = 50, r = 15, b = 10, l = 15)
      )
```

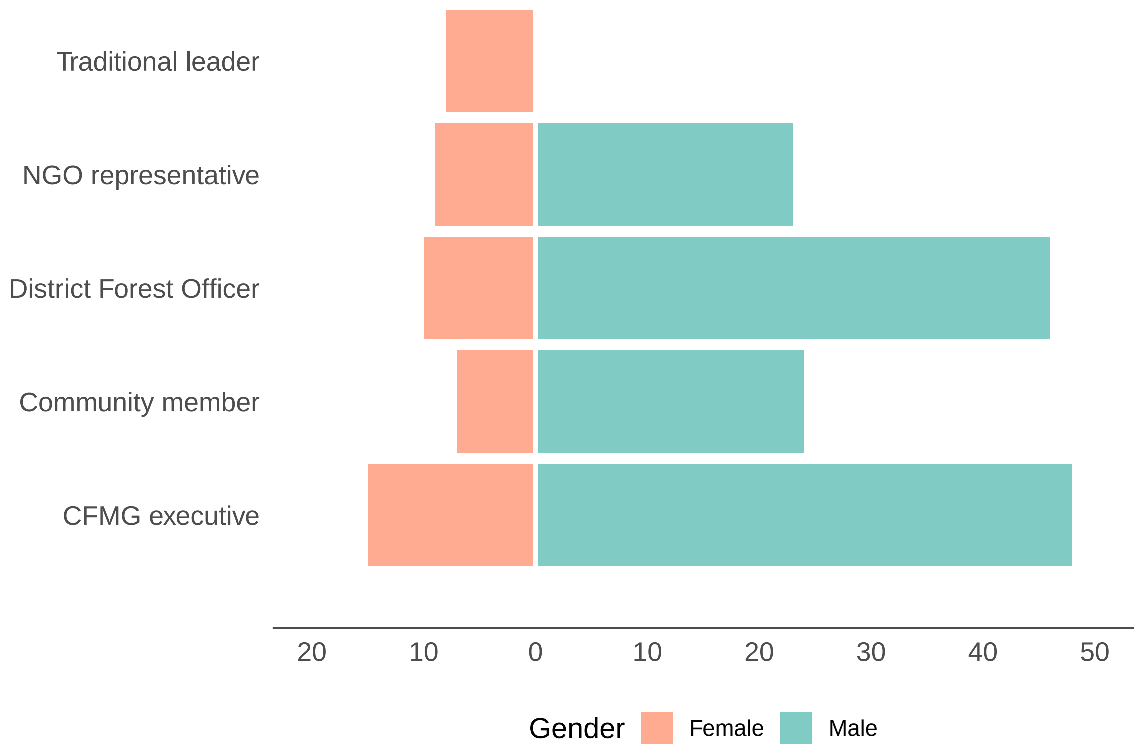

Figure 6: Gender breakdown of individual interviews.

Age breakdown of intrerviewees

The ages of interviewees were broadly similar among men and women (Figure 7). While the median age of individual interviewees was slightly lower among men, the mean age was slight higher. This arises because the male sample included some much older men.

```
cfmg |> filter(!is.na(demo_information_interviewee_gender)) |>
        ggplot(aes(x = demo_information_interviewee_gender, y = demo_information_interviewee_
        geom_boxplot() +
        geom_jitter()+
        stat_summary(fun=mean, geom="point", size=3, col = 'red') +
        scale_fill_manual(
    values = c("female" = "#FFAB91", "male" = "#80CBC4"),
    labels = c("Female", "Male"),
```

```
    name = "Gender"
  ) +
        scale_y_continuous(
                limits = c(20,90)
        ) +
        theme_bw(base_size = 14) +
        theme(legend.position = "none",
              axis.text.y = element_text(size = rel(1.14)),
              axis.text.x = element_text(size = rel(1.14), angle = 0, vjust = 1, hjust=0.5),
              axis.title = element_text(size = rel(1.29))) +
        xlab("") +
        ylab("Age of interviewees")
```

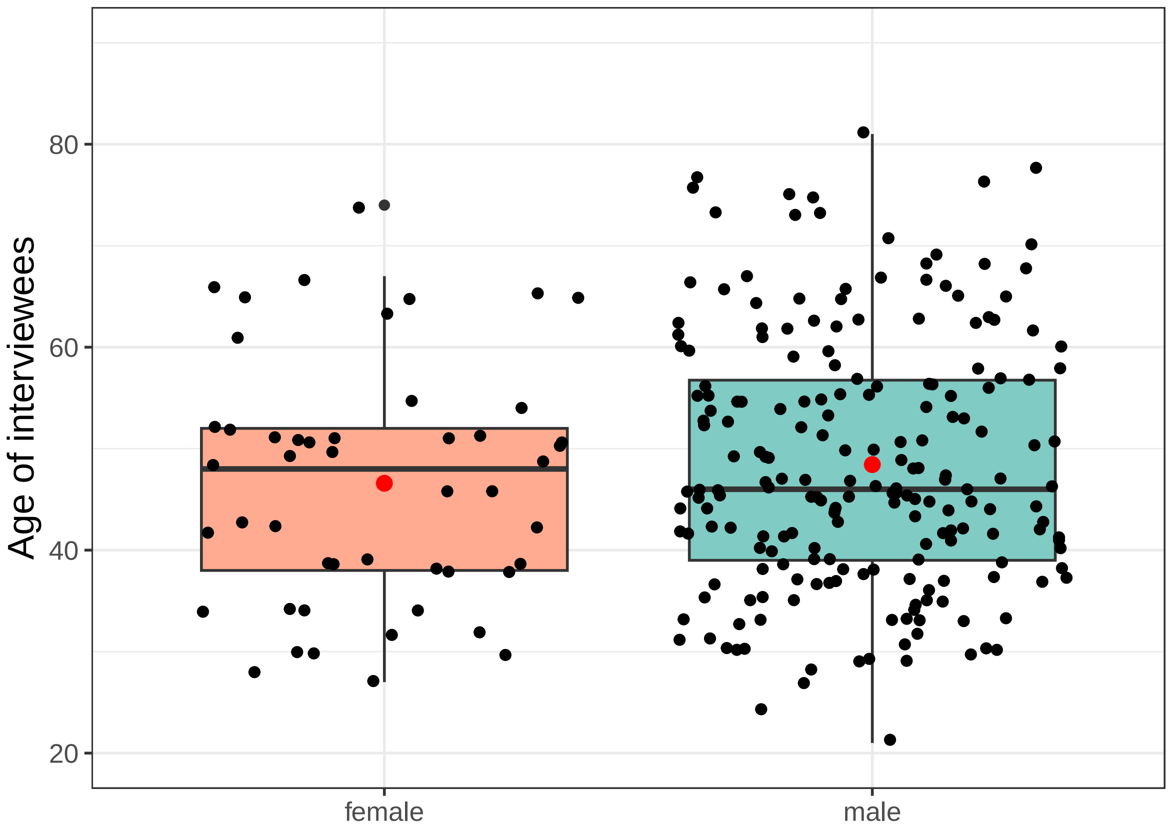

Figure 7: The age of individual interviewees by gender. The black dots represent individual interviewees. The box encompasses the 25th to 75th percentile with the 50th percentile denoted by the black bar. The red dot represents the mean.

Years interviewees have lived in their communities

When we examine the number of years living in the community (Figure 8), we find that the men interviewed had on average lived in the community longer than the women interviewed. In addition, the results reveal that some people interviewed had moved into the community in recent years.

```
cfmg |> filter(!is.na(demo_information_interviewee_gender)) |>
        ggplot(aes(x = demo_information_interviewee_gender,
                   y = demo_information_interviewee_years_living_comm,
                   fill = demo_information_interviewee_gender)) +
        geom_boxplot() +
        geom_jitter()+
        stat_summary(fun=mean, geom="point", size=3, col = 'red') +
        scale_fill_manual(
    values = c("female" = "#FFAB91", "male" = "#80CBC4"),
    labels = c("Female", "Male"),
    name = "Gender"
  ) +
        scale_y_continuous(
                limits = c(0,80)
        ) +
        theme_bw(base_size = 14) +
        theme(legend.position = "none",
              axis.text.y = element_text(size = rel(1.14)),
              axis.text.x = element_text(size = rel(1.14), angle = 0, vjust = 1, hjust=0.5),
              axis.title = element_text(size = rel(1.29))) +
        xlab("") +
        ylab("Years in community")
```

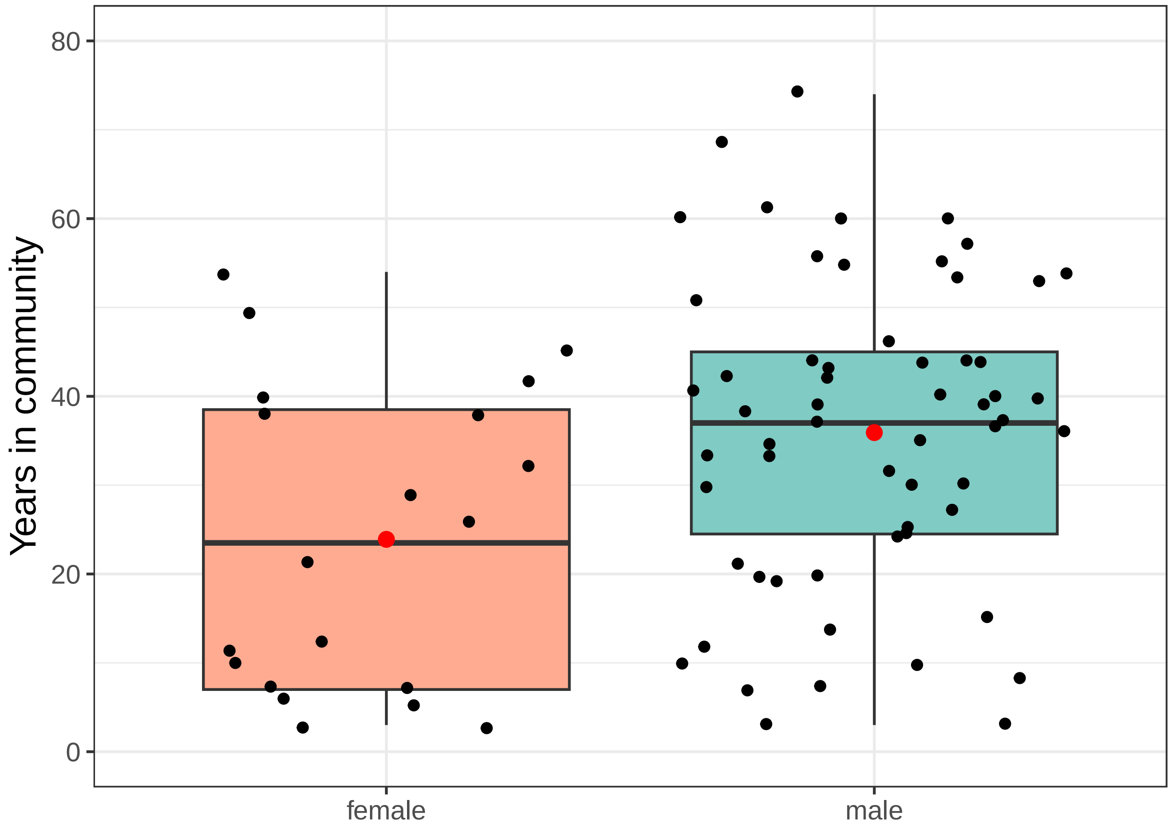

Figure 8: The number of years individual interviewees claimed to have lived in the community by gender. The black dots represent individual interviewees. The box encompasses the 25th to 75th percentile with the 50th percentile denoted by the black bar. The red dot represents the mean.

Looking at the same information by interviewee role (Figure 9), we see that both the CFMG executive members interviewed and the community members interviewed included a wide range of participants from those who had recently moved into the community to those that had lived there for 50 yrs or more.

```
cfmg |> filter(interviewee_role == "cfmg_member" |
                     interviewee_role == "cfmg_exec_member" ) |>
        mutate(
                interviewee_role = case_when(
                        interviewee_role == "cfmg_member" ~ "Community member",
                        interviewee_role == "cfmg_exec_member" ~ "CFMG executive")
                ) |>
```

```
ggplot(aes(x = interviewee_role,
           y = demo_information_interviewee_years_living_comm,
           )) +
geom_boxplot() +
geom_jitter()+
stat_summary(fun=mean, geom="point", size=3, col = 'red') +
scale_fill_manual(
        values = c("CFMG executive" = "#FFAB91",
                   "Community member" = "#80CBC4"),
        labels = c("CFMG executive", "Community member"),
        name = "Interviewee Role"
        ) +
scale_y_continuous(
        limits = c(0,80)
) +
theme_bw(base_size = 14) +
theme(legend.position = "none",
      axis.text.y = element_text(size = rel(1.14)),
      axis.text.x = element_text(size = rel(1.14), angle = 0, vjust = 1, hjust=0.5),
      axis.title = element_text(size = rel(1.29))) +
xlab("") +
ylab("Years in community")
```

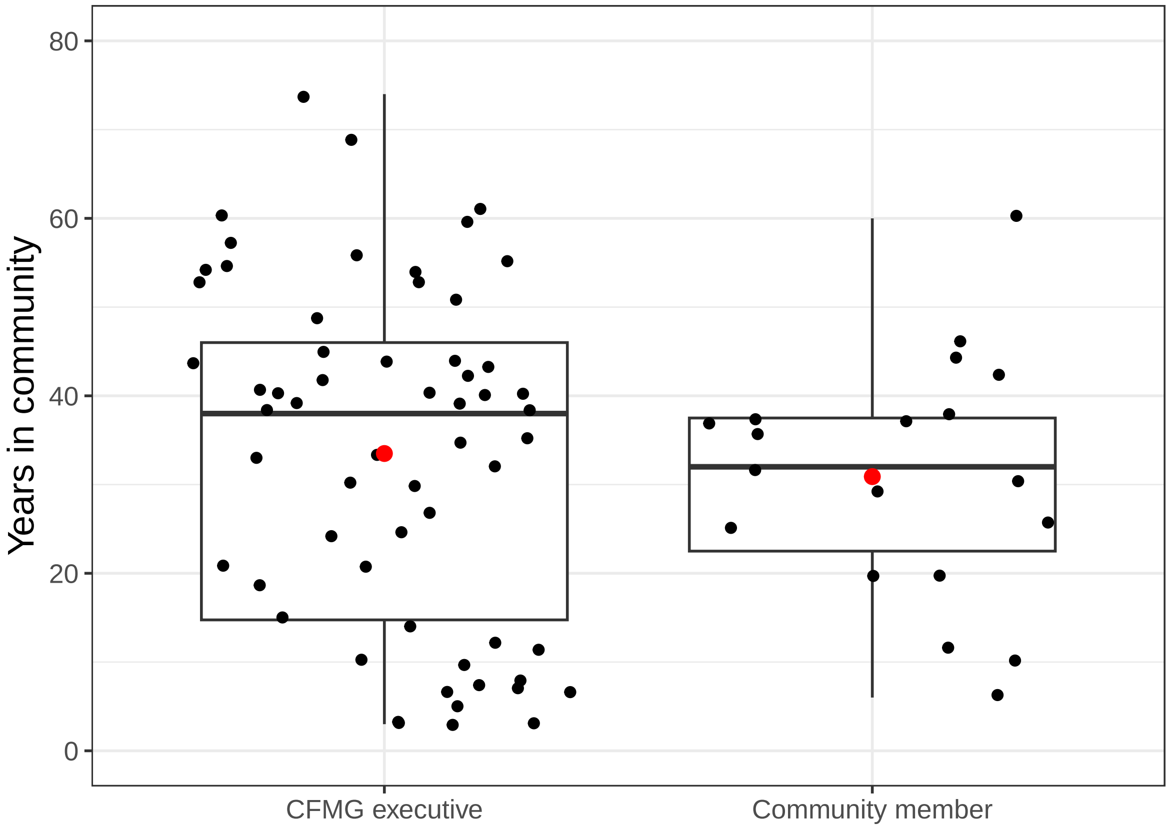

Figure 9: The number of years individual interviewees claimed to have lived in the community by interviewee role. The black dots represent individual interviewees. The box encompasses the 25th to 75th percentile with the 50th percentile denoted by the black bar. The red dot represents the mean.

Duration of involvement with CFMG

We asked the interviewees how many years they had been involved in the CFMG (Figure 10). The answers ranged from zero to ten years with the median among provinces ranging from around 3 yrs to 4.5 yrs. As the regulations for community forest management were only approved in 2018, claims of over 7 yrs seem unlikely but they may have been involved in a Community Resource Board (CRB) or in Joint Forest Management (JFM), which preceded CFM in some areas. As most CFMGs have only held one election as part of their establishment process, we can expect that most people have been involved since the formation of their CFMG.

```r
cfmg |> filter(!is.na(demo_information_interviewee_years_involved_cfmg)) |>

        ggplot(aes(x = province,
                   y = demo_information_interviewee_years_involved_cfmg)) +
        geom_boxplot() +
        geom_jitter(width = 0.1) +
        stat_summary(fun=mean, geom="point", size=3, col = 'red') +
        theme_bw(base_size = 14) +
        theme(legend.position = "none",
              axis.text.x = element_text(size = rel(1.14), angle = 45, hjust=1),
              axis.text.y = element_text(size = rel(1.14)),
              axis.title.y = element_text(size = rel(1.29))) +
        scale_y_continuous(
                limits = c(0,10)
        ) +
        xlab("") +
        ylab("Years involved in CFMG")
```

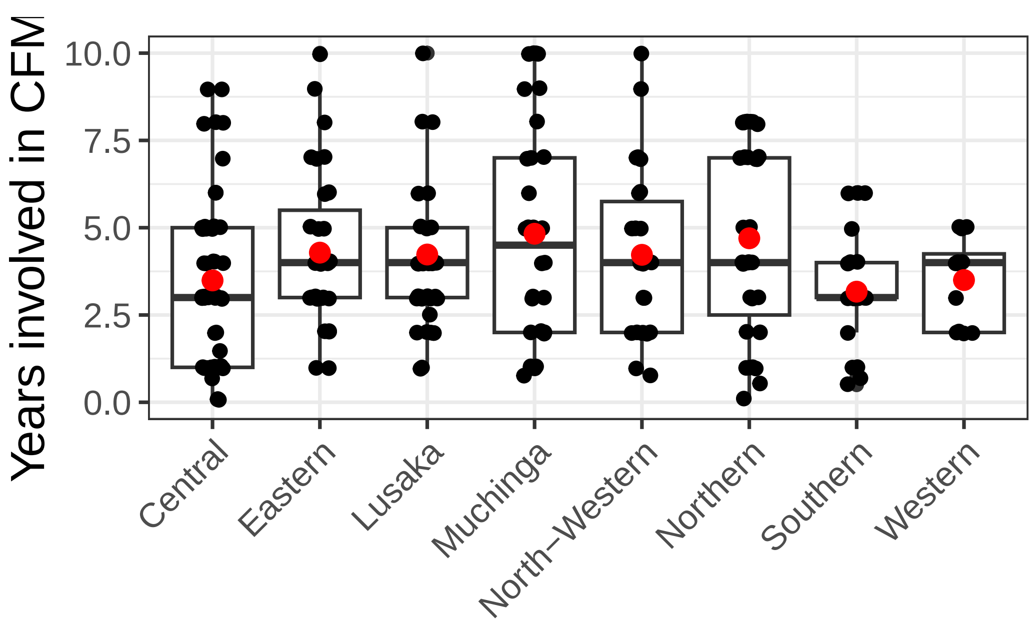

Figure 10: Number of years interviewees claimed to be involved in the CFMG by province. The black dots represent individual interviewees. The box encompasses the 25th to 75th percentile with the 50th percentile denoted by the black bar. The red dot represents the mean.

Discussion

This chapter describes the sample of community forests and interviewees that form the basis of all subsequent analyses in this report. The central question for this chapter is whether the sample is sufficiently unbiased and broad to support inferences about CFM implementation at a national level in Zambia.

The selection of CFs followed a stratified random design: CFs were randomly selected within each of eight provinces, with a minimum age threshold of one year at the time of sampling. The two provinces omitted — Luapula and Copperbelt — had very few CFs at the time of the survey, and therefore their exclusion does not compromise national representativeness. Five CFs included in the original sample could not be surveyed after it became clear they existed only on paper. The final sample of 75 CFs is provided in Appendix II. The sample spans a wide range of CF sizes and ages: from very small CFs of around 20 ha to large ones exceeding 80,000 ha, and from newly established CFs of one year to approximately seven years old. Because the CFM programme only received enabling regulations in 2018, most CFs in Zambia are fewer than three years old, and our sample reflects this age structure appropriately. Together, the geographic and structural breadth of the sample supports the conclusion that it is representative of the national CF population.

The multi-perspective FGD design is the key methodological feature that enables meaningful inference from interview data. By conducting four structured FGDs per CF — with the CFMG executive committee, community men, community women, and community youth — the survey is able to capture systematically divergent perspectives within each community. The near-complete coverage of all four FGD types across the 75 CFs means that differences in responses among these groups, which feature prominently throughout the report, can be interpreted with confidence.

Individual interviews were targeted at five key stakeholder roles: CFMG executive members, DFOs, traditional leaders, forest user-group chairs, and external partners. Not all roles were available at every CF, which introduces some variation in coverage, but the most important roles — CFMG executives, DFOs, and traditional leaders — were represented at the large majority of sites.

The gender breakdown of interviewees reflects real-world patterns: men dominate most CFMG stakeholder roles, and so the overall sample is male-biased. However, the dedicated female FGD in each CF and the inclusion of female individual interviewees where available ensure that women's perspectives are not lost. The age structure and 'years living in the community' of interviewees suggest a broadly representative sample of community members, spanning both recent arrivals and long-term residents across all provinces. The years of CFMG involvement reported by interviewees are consistent with the age of the programme, and where involvement exceeds seven years it likely reflects participation in predecessor community resource management structures, such as CRBs.

In conclusion, the stratified random selection of CFs, combined with the structured multi-perspective FGD design and the diversity of individual interviewee roles, gender, age and experience, provides a robust foundation for national-level inference about CFM implementation in Zambia.